$$\frac{\cancel{5}\times 10^{-3}}{\cancel{5}} \cdot 13\times 10^{3} - 9$$

$$1\times \cancel{10^{-3}} \cdot 13\times \cancel{10^{3}} - 9$$

$$13 - 9 = 4$$

(3)

$$\left(\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{\sqrt{16^{32}}}}}}}}}}}}}}}}\right)^{3} - 4 =$$

$$\left(\sqrt{\sqrt{16}}\right)^{3} - 4 =$$

$$\left(\sqrt{4}\right)^{3} - 4 =$$

$$2^{3} - 4 =$$

$$8 - 4 = 4$$

(4)

$$\sqrt[9]{3\log_{10}\left[2\cdot\left(5\cdot\frac{\log_4 64}{\log_3 27}\right)\right]^{2}} - 23 =$$

$$\sqrt[9]{3\log_{10}\left[2\cdot(5\cdot 2)\right]^{2}} - 23 =$$

$$\sqrt[9]{3\log_{10} 20^{2}} - 23 =$$

$$9\cdot 3\log(10+10) - 23 =$$

$$9\cdot 3\cdot 1\cdot 1 - 23 =$$

$$27 - 23 = 4$$

(5)

$$-5 + \left(\frac{a^{2}b^{2}c}{ab\sqrt{c}} + \frac{abc^{4}}{c^{2}\sqrt{c}} - \frac{ab^{5}c^{2}}{b^{4}a\sqrt{c}}\right)\cdot\frac{9}{ab\sqrt{c}}$$

$$-5 + \left(\frac{\cancel{abc}}{\sqrt{c}} + \frac{\cancel{abc}}{\cancel{\sqrt{c}}} - \frac{\cancel{abc}}{\cancel{\sqrt{c}}}\right)\cdot\frac{9}{\cancel{ab\sqrt{c}}}$$

$$-5 + \frac{c}{\sqrt{c}}\cdot\frac{9}{\sqrt{c}}$$

$$-5 + \frac{9c}{\sqrt{c^{2}}} \rightarrow -5 + \frac{9\cancel{c}}{\cancel{c}} = 4$$

$$\frac{\log_4 64}{\log_3 27}$$

$$\frac{\log 64}{\log 4}$$

$$\frac{\log 27}{\log 3}$$

$$\frac{\log 64 - \log 4}{\log 27 - \log 3}$$

$$\frac{\cancel{\log 2^{6}} - \cancel{\log 2^{2}}}{\cancel{\log 3^{3}} - \cancel{\log 3^{1}}} = \frac{6-2}{3-1} = \frac{4}{2} = 2$$